

Real Estate Management System

Team Members :

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Contributions :

Vighnesh :

1. Designed the overall ER model and defined primary keys.
2. Ensured the schema matches project requirements.
3. Contributed to normalization of the database.

Prabhat :

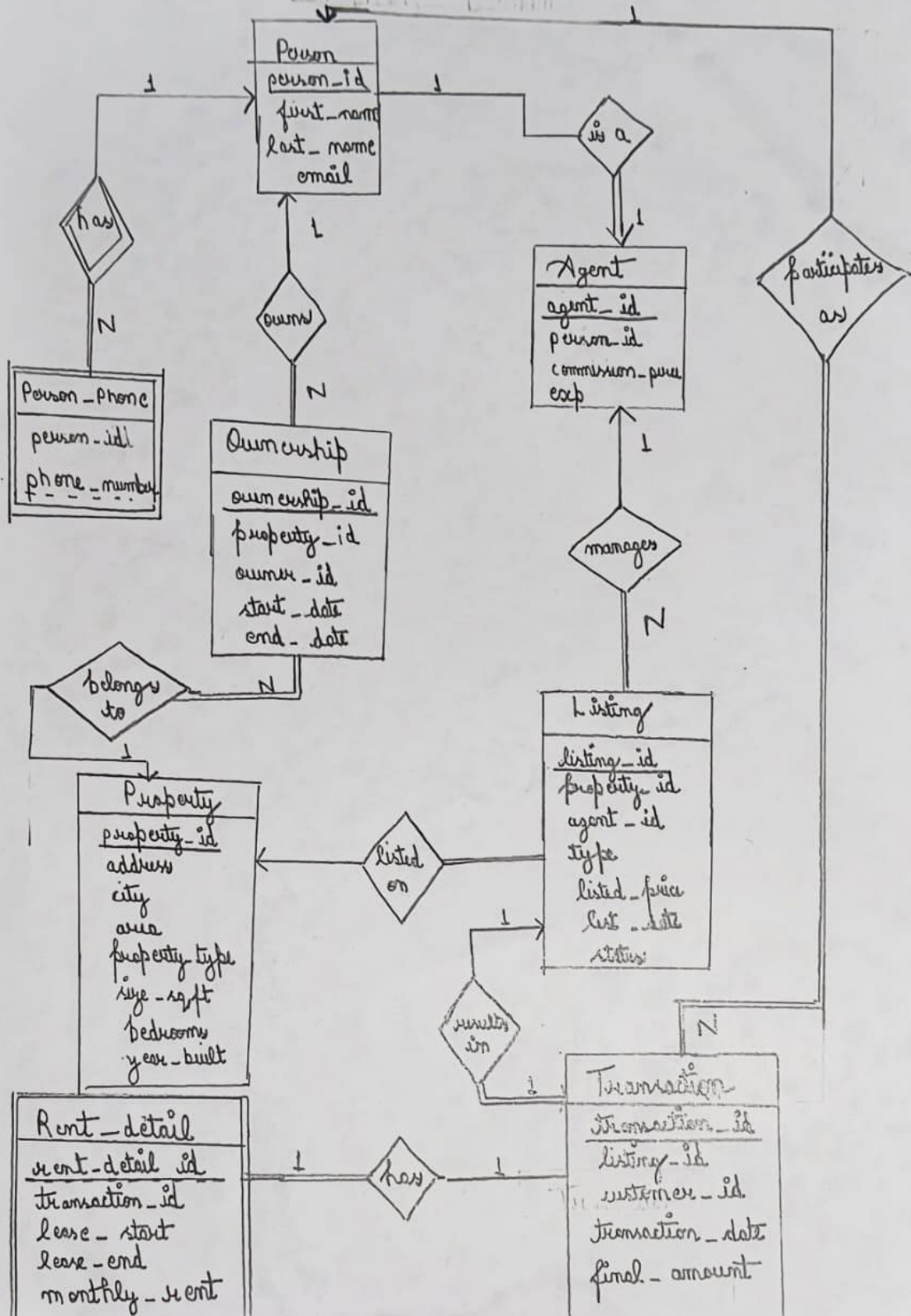
1. Identified main entities and their attributes.
2. Reviewed the ER model and validated assumptions.
3. Contributed to normalization and tested queries to improve model relevance.

Pradyumn :

1. Verified correctness of relationships in the ER model.
2. Refined structure of the diagram.
3. Contributed to normalization.

AI usage: AI was used to verify whether the initial ER diagram satisfied all the requirements of the question, and minor changes were made based on those suggestions.

ER Diagram



Assumptions :

1. A single PERSON table is used to represent all individuals (agents, buyers, owners, renters), with roles determined by their relationships in other tables.
2. Each person can have at most one agent profile, enforced by a UNIQUE constraint on AGENT.person_id.
3. Each listing is associated with exactly one agent; co-listing or multiple agents per listing are not supported.
4. Each listing can result in at most one transaction, enforced by a UNIQUE constraint on TRANSACTION.listing_id.
5. Ownership is modeled as a temporal relationship using the OWNERSHIP table, with start_date and end_date, where end_date = NULL represents the current owner.
6. Ownership changes only in case of a sale transaction; rental transactions do not affect property ownership.
7. Listings are not deleted from the system; instead, their lifecycle is managed using a status attribute (active, sold, rented, withdrawn).
8. A listing is used only once; after a transaction, a new listing must be created for future sales or rentals of the same property.
9. Property types are restricted to predefined values (house, apartment) using a CHECK constraint.
10. The transaction date must be on or after the listing date, enforced through a database trigger.

Triggers

1. trg_validate_transaction_date ensures that a transaction date cannot be earlier than the listing date.
2. trg_validate_rent_detail prevents adding rent details for sale-type transactions.
3. trg_after_transaction_complete automatically updates listing status and ownership after a transaction.
4. trg_prevent_duplicate_active_listing blocks creating multiple active listings for the same property.
5. trg_validate_ownership_on_listing prevents listing a property that has no owner.
6. trg_validate_lease_dates ensures that the lease end date is after the lease start date.

Diagram

